

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

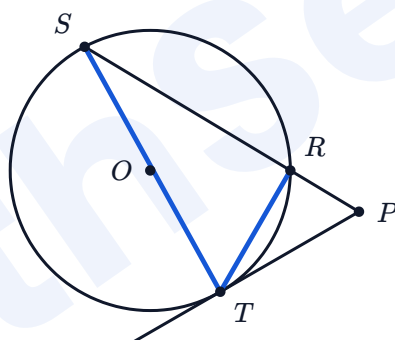
Chapter 18: Tangents and Intersecting Chords

Previous year board questions, topic-wise: 2 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

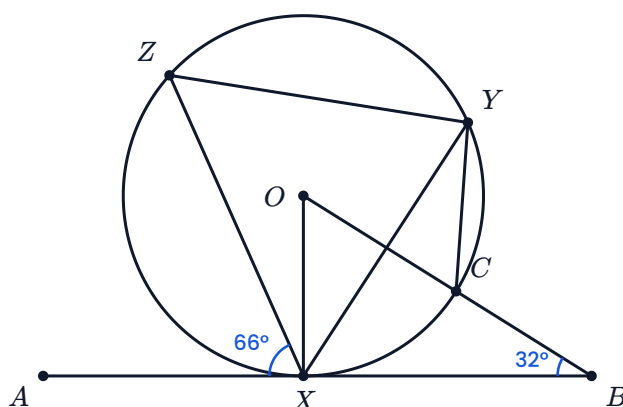
1 Tangent and secant from an external point

1. PT is a tangent to a circle at T , and a line through P cuts the circle at R and S , with R nearer to P . (a) Prove that $\triangle PTR \sim \triangle PST$. (b) Prove that $PT^2 = PR \times PS$. (c) If $PR = 4$ cm and $PS = 16$ cm, find PT . **ICSE 2025, 3 marks**



2 The alternate segment theorem

2. X, Y, Z and C lie on a circle with centre O . AB is the tangent at X , OB cuts the circle at C , and $ZY = XY$. Given $\angle OBX = 32^\circ$ and $\angle AXZ = 66^\circ$, find (a) $\angle BOX$ (b) $\angle CYX$ (c) $\angle ZYX$ (d) $\angle OXY$. **ICSE 2025, 4 marks**



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